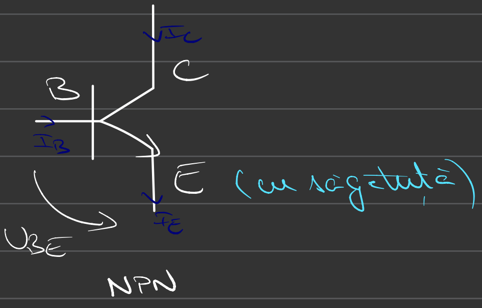
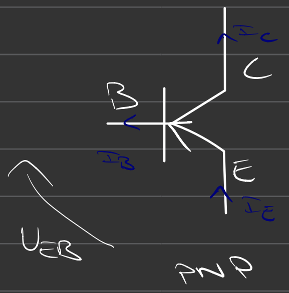
Seminar 4





RAN : $I_C = \beta I_B$

$U_{BE} = 0{,}6$

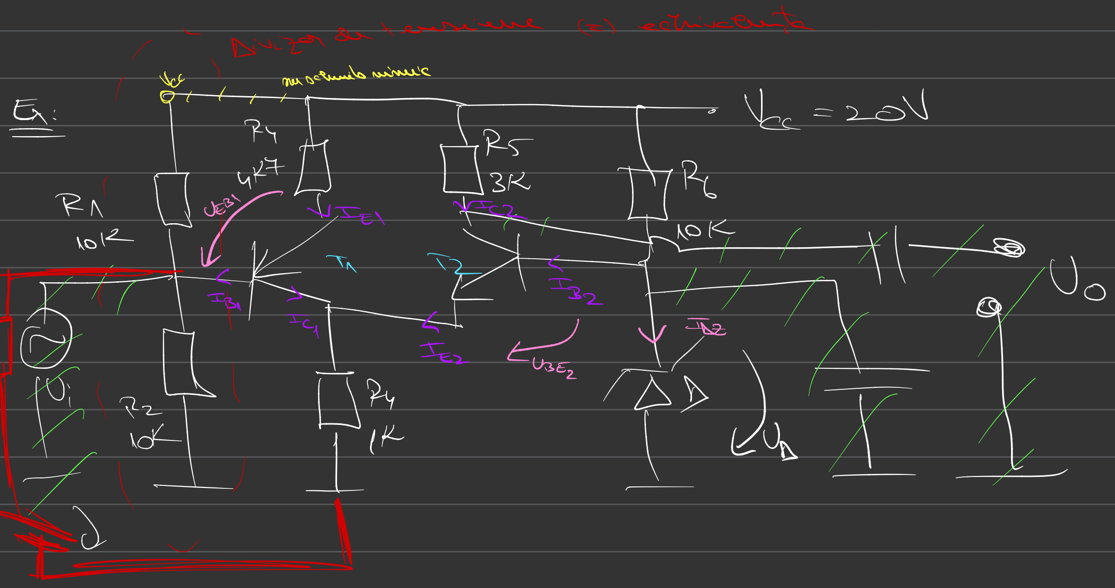
$$\boxed{I_B + I_C = I_E}$$

$U_{EB} = 0{,}6$

$I \simeq I_C$

$U_{CE} > U_{BE}\,(0{,}6V)$

$U_{EC} > U_{EB}\,(0{,}6V)$



avem Rc și Ec

$U_{DZ} = 3V6$

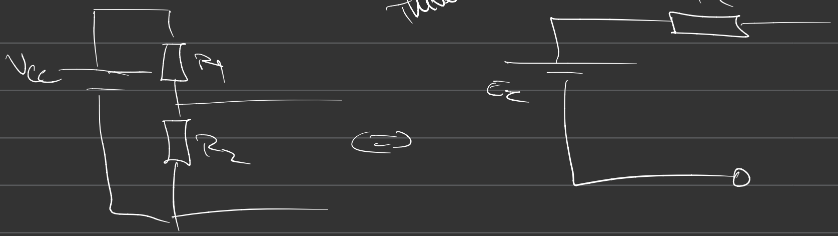
P.S.F.: I_C, U_{CE}

Pp că: $I_D > 0 \Rightarrow U_D = U_{DZ} = 3{,}6\,V$

$R_e = R_1 \| R_2 = 5K$

$$E_e = \frac{(V_{CC} - 0) R_2}{R_1 + R_2} = \frac{20 \cdot 10}{20} = 10\,V$$

echivalenta Thevenin



E_e → echivalenta de mers în gol

R_e → R circuitului pasivizat

Trec prin BAZA + EMITOR ⇒ Aflu intensitatea

$$\Rightarrow \quad V_{cc} = I_{E1} \cdot R_3 + U_{EB} + I_{B1} \cdot R_E + E_e$$

$$V_{cc} = I_{C1} \cdot R_3 + U_{EB} + \frac{I_C}{\beta} \cdot R_E + E_e$$

$$I_{C1} = \frac{V_{cc} - U_{EB} - R_E}{\frac{R_E}{\beta} + R_3} = \frac{20 - 0{,}6 - 10}{4K1 + \frac{5K}{100}}$$

$$I_{C1} = 2mA$$

$$U_A = U_{BE2} + R_4(I_{C1} + I_{E2})$$

$$\Rightarrow I_{C2} = 1mA$$

$V_{CC} = I_{E1} R_3 + U_{EC1} + R_4 (I_{C1} + I_{E2})$

$V_{CC} = I_{C1} R_3 + U_{EC1} + R_4 (I_{C1} + I_{E2})$

$U_{EC1} = V_{CC} - \underset{2mA \;\; 4K7}{I_{C1} R_3} - \underset{2mA \;\; 1mA}{R_4 (I_{C1} + I_{C2})}$

$= 20 - 9{,}4 - 3 \quad = \quad 10{,}6 - 3 = 7{,}6 = \underline{7{,}6V} > 0{,}6V$

Adev ⇒ RAN

$V_{CC} - 0 = I_{C2} R_5 + U_{CE2} + R_4 (I_{C1} + I_{E2})$

$U_{CE2} = 20 - 3 - 3$

$U_{CE2} = 14\ V > 0{,}6$ ⇒ Adev ⇒ RAN

Presupunerea de la început e adev?

$I_D > 0$

$V_{CC} = U_6 + U_D \iff 20 = I_6 R_6 + 3{,}6$

$\frac{16{,}4}{10K} = I_6 \Rightarrow \underline{I_6 = 1{,}64\ mA}$

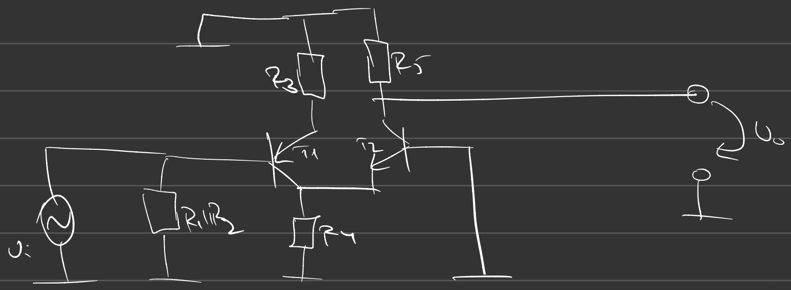
$I_D + I_{B2} = I_6 \iff 1{,}64 = I_D + 1$

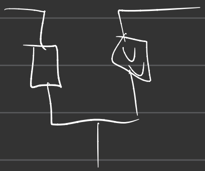
$0{,}64 = I_D > 0$ Adev

Dacă nu foloseam echivalenta E_e, R_e: Aplicam sistem K:

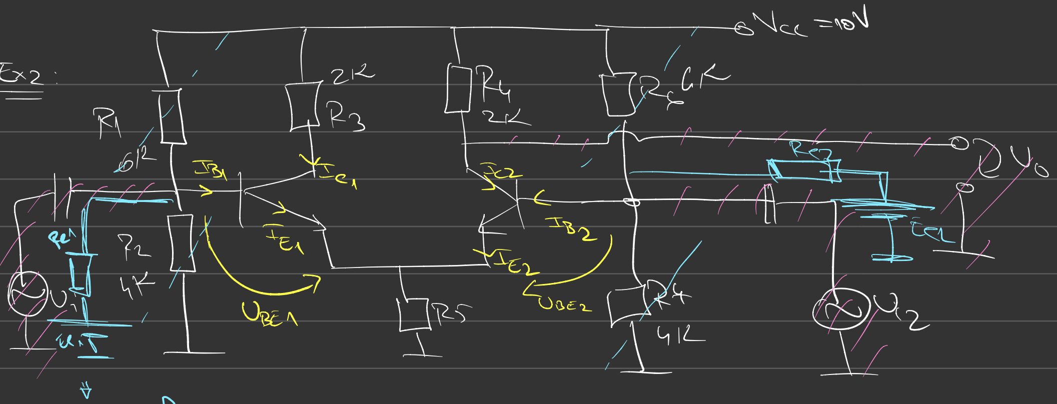
$$\begin{cases} I_1 + I_{B1} = I_2 \\ V_{CC} = I_{E1} R_3 + U_{EB1} + I_2 R_2 \\ I_1 R_1 = U_{EB1} + I_{E1} \cdot R_3 \end{cases}$$

Ex: Redesenhar





Ex2:



⇓

echivalenta ?

Amplificator diferential   P.S.F. I_{C1}, U_{CE}

$R_{E1} = R_1 \parallel R_2 = \frac{6\cdot 4}{10} = 2{,}4\ k\Omega$

$E_{e1} = \frac{(V_{cc} - 0)\cdot R_2}{R_1 + R_2} = \frac{10\cdot 4}{10} = 4V$

$R_{E2} = 2{,}4\ k\Omega$

$E_{e2} = 4V$

Schema simetrica $\Rightarrow I_{C1} = I_{C2}$ (sunt egali)

$U_{CE1} = U_{CE2}$

$0 = I_{B1}\cdot R_{E1} + U_{BE1} + 2 I_{E1}\cdot R_5 - E_{e1}$

$0 = \frac{I_{C1}}{\beta} R_{E1} + U_{BE1} + 2 I_{C1}\cdot R_5 - E_{e1}$

$I_{C1} = \frac{E_{e1} - U_{BE1}}{\frac{R_{E1}}{\beta} + 2R_5} \Leftrightarrow I_{C1} = 1\ mA$

$V_{CC} = I_{C1} R_3 + U_{CE} + 2 I_E R_5$

$U_{CE} = V_{CC} - I_C (R_3 + 2R_5)$

$U_{CE} = 10 - 1 \cdot 5{,}4 \Rightarrow U_{CE} = 4{,}6$ ✓ $\geq 0{,}6$ Adev

$\Rightarrow$ RAN